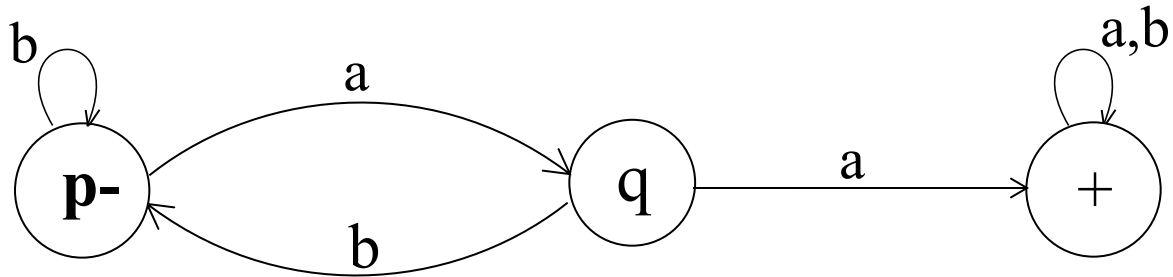


# Recap Lecture 17

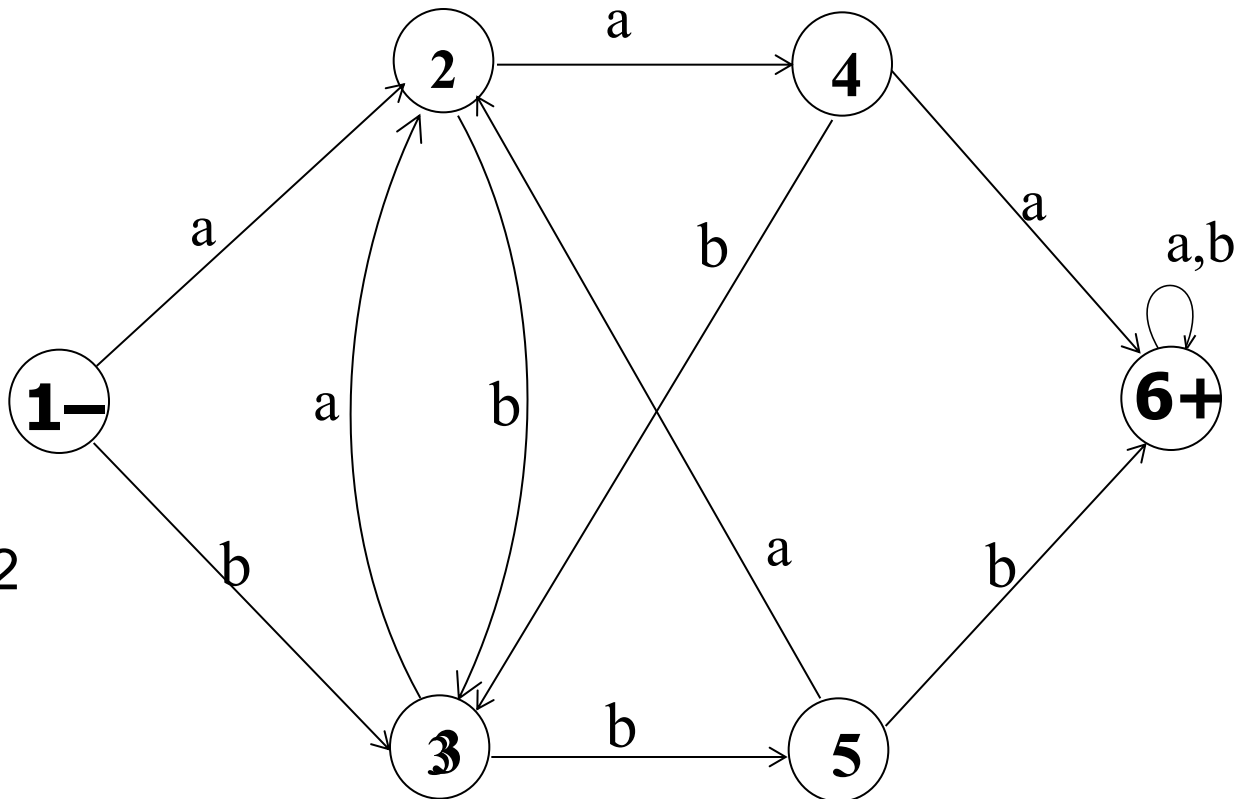
- converting NFA to FA (method 3), example, NFA and Kleene's theorem method 1, examples, NFA and Kleene's theorem method 2 , NFA corresponding to union of FAs, example

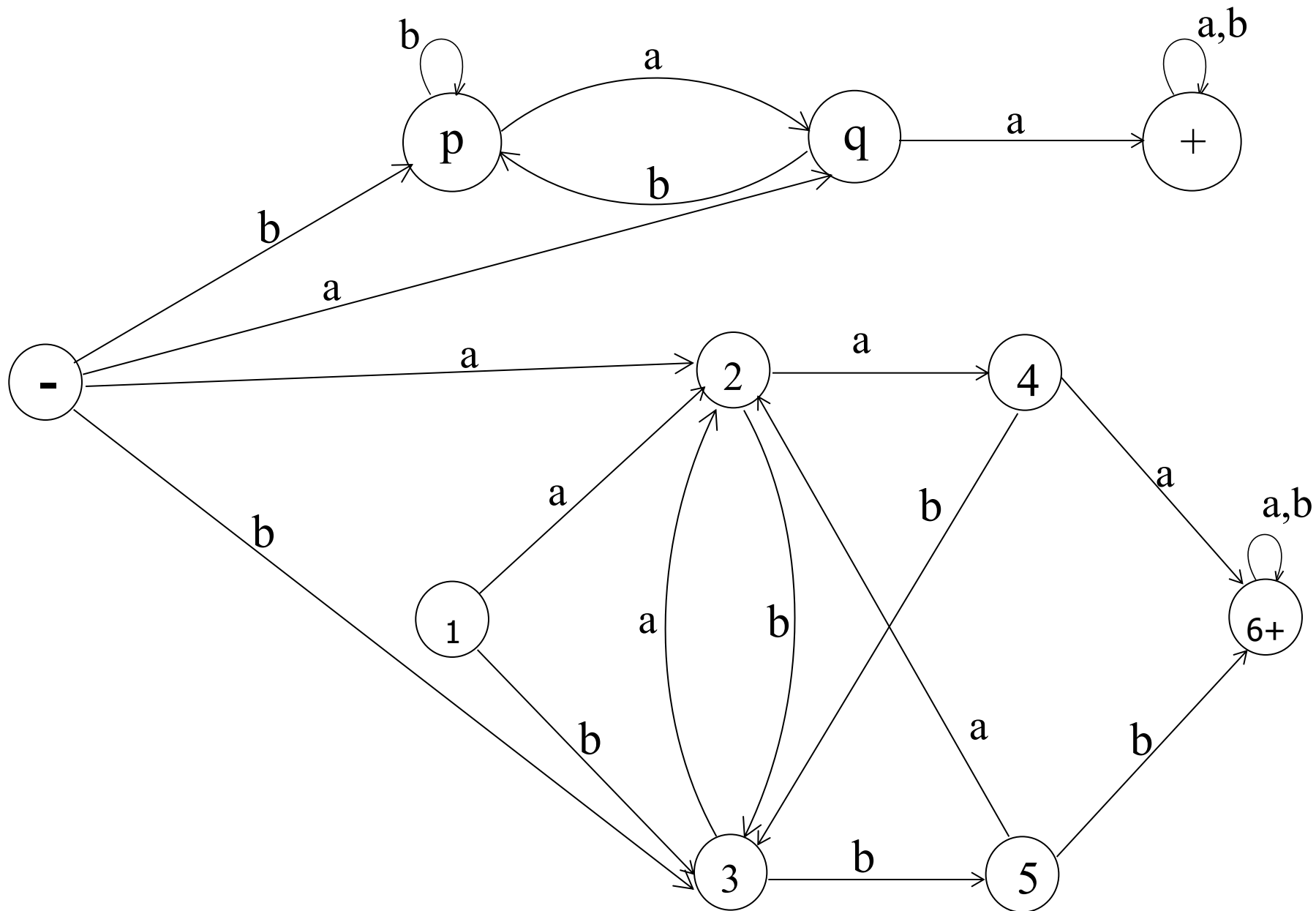
# Example

- $FA_1$



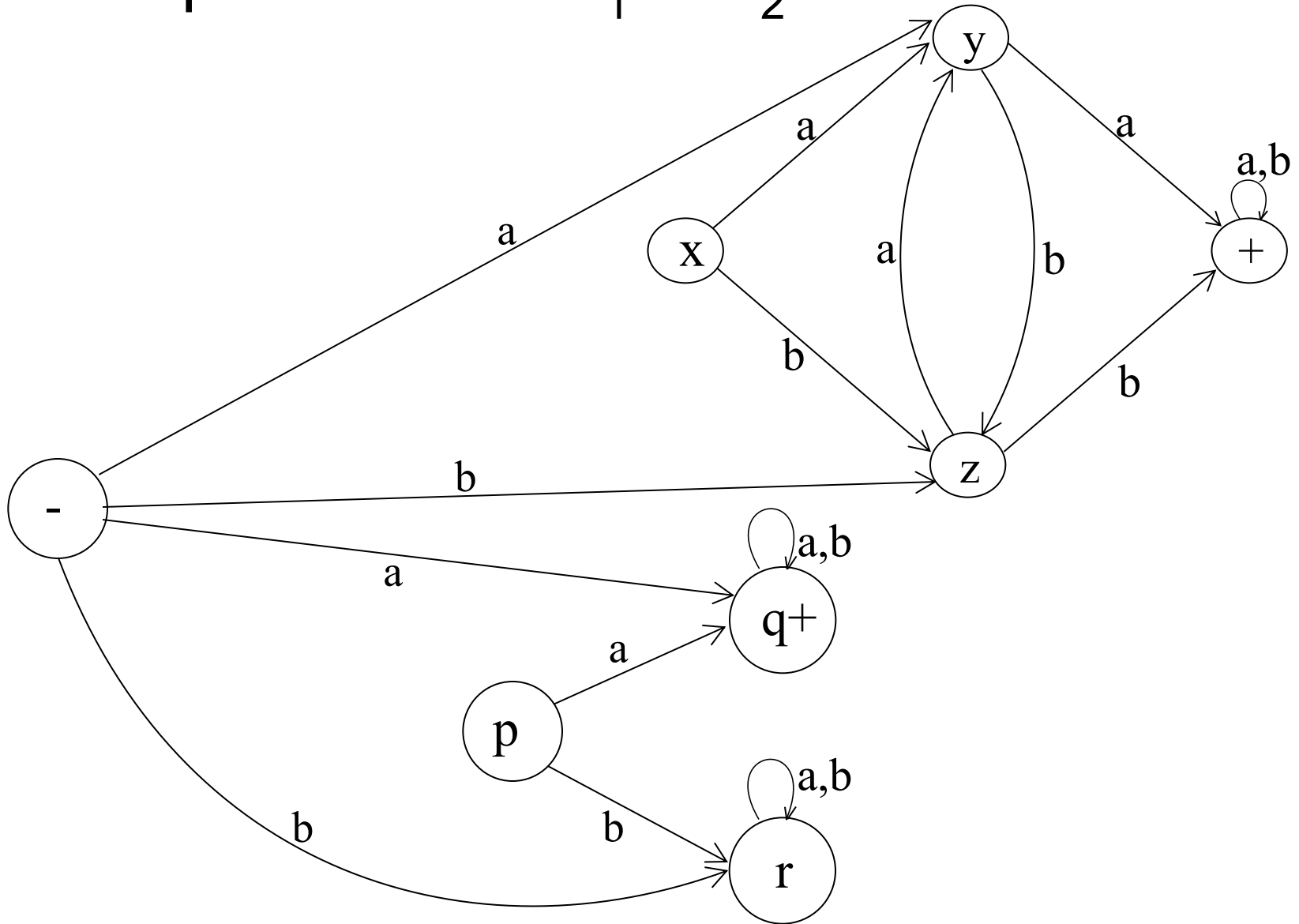
- $FA_2$





NFA equivalent to  $FA_1 UFA_2$

# NFA equivalent to $FA_1 U FA_2$



# NFA corresponding to Concatenation of FAs

## Method:

Introduce additional transitions for each letter connecting each final state of the first FA with the states of second FA that are connected with the initial state of second FA corresponding to each letter of the alphabet. Remove the +ve sign of each of final states of first FA and –ve sign of the initial state of second FA. It will create non-determinism at final states of first FA and hence NFA, thus obtained, will be the required NFA.

# NFA of Concatenation of FAs

## Continued ...

### **Note:**

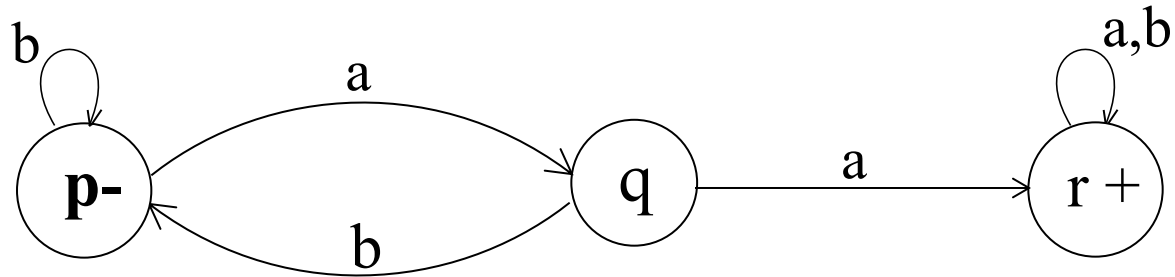
It may be noted that if first FA accepts the Null string then every string accepted by second FA must be accepted by the concatenation of FAs as well. This situation will automatically be accommodated using the method discussed earlier. However if the second FA accepts Null string, then every string accepted by first FA must be accepted by the required FA as well. This target can be achieved as, while introducing new transitions at final states of first FA the +ve sign of these states will not be removed.

## Note Continued ...

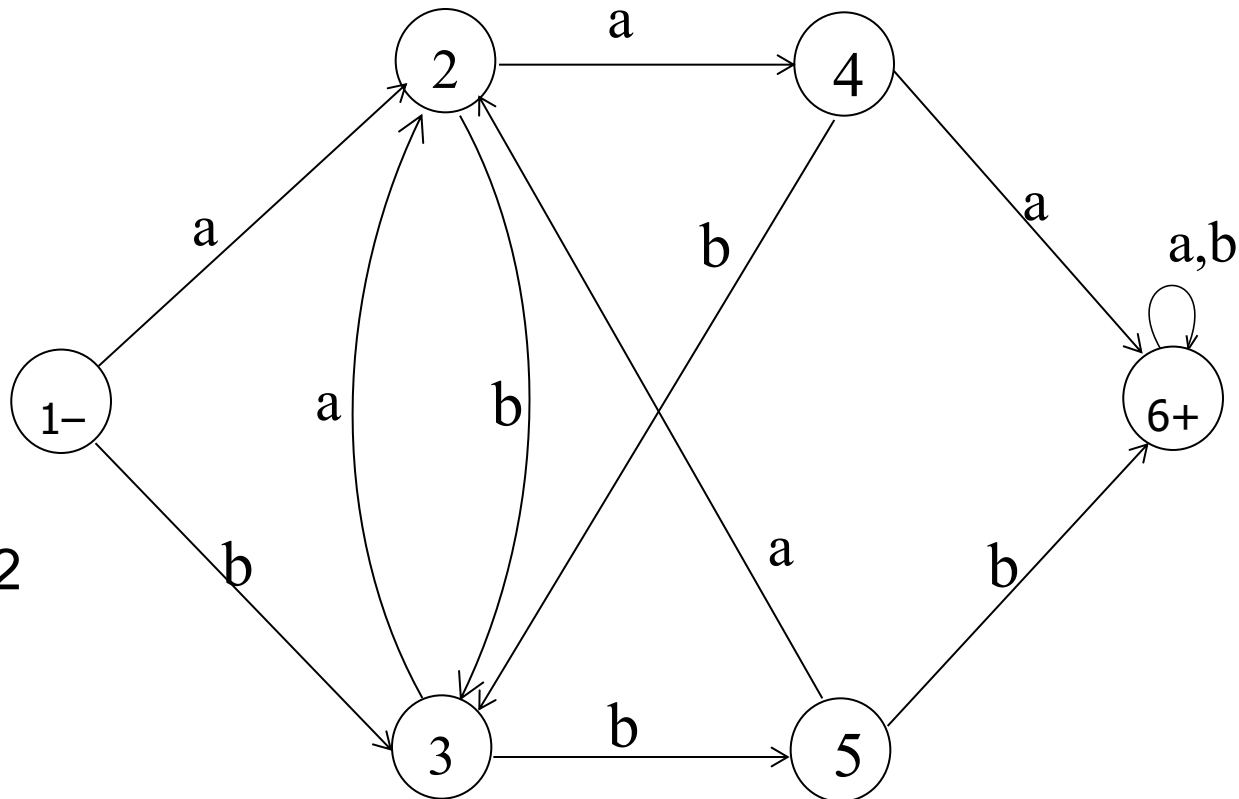
Lastly if both FAs accept the Null string, then the Null string must be accepted by the required FA. This situation will automatically be accommodated as the second FA accepts the Null string and hence the +ve signs of final states of first FA will not be removed.

# Example (No FA accepts Null string)

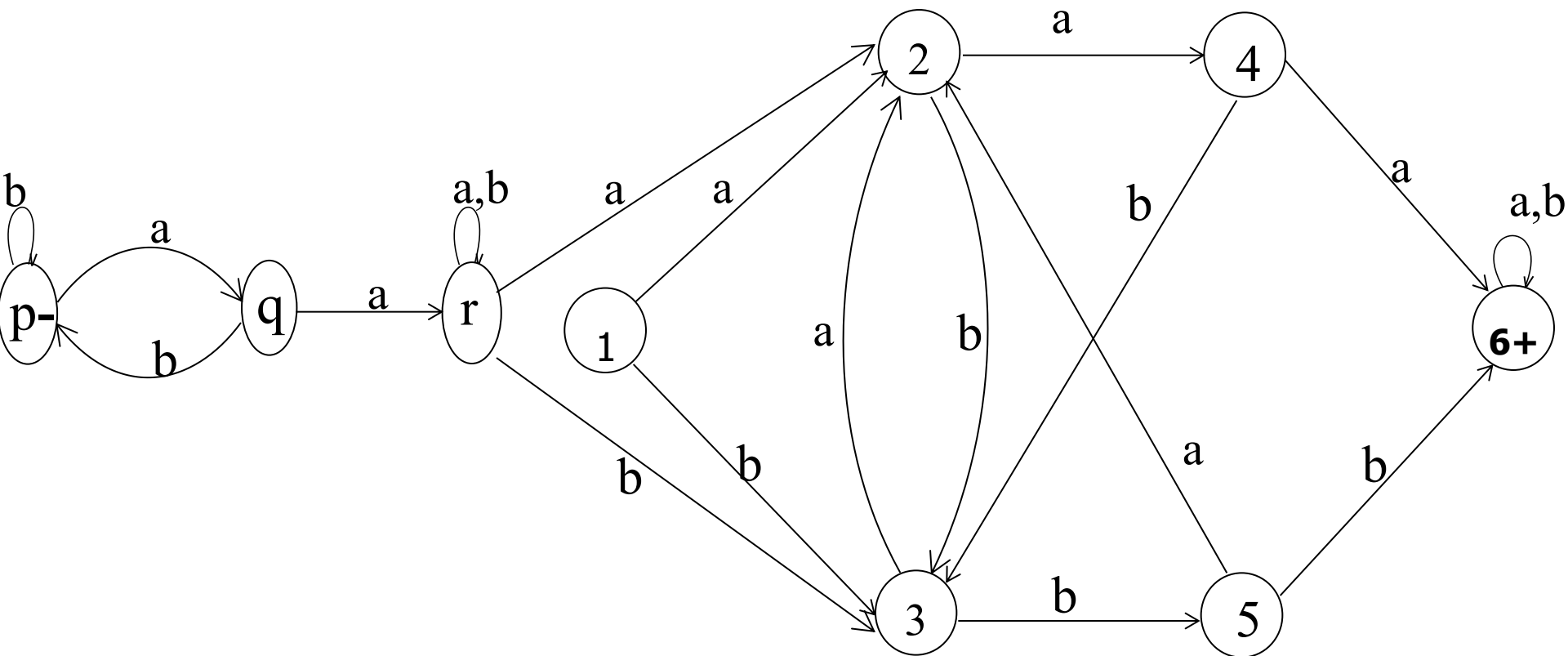
- $FA_1$



- $FA_2$



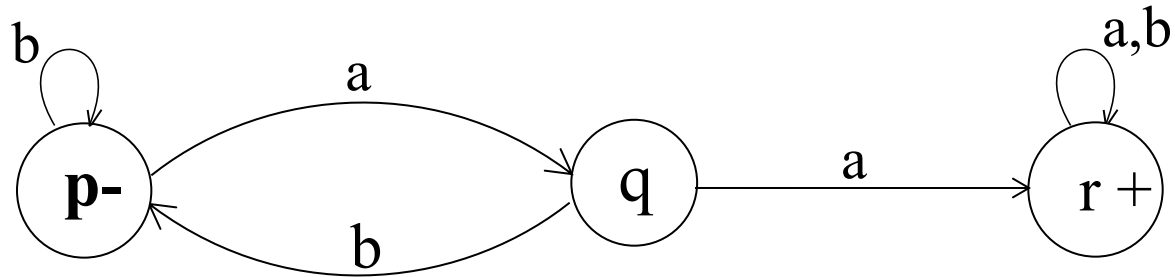




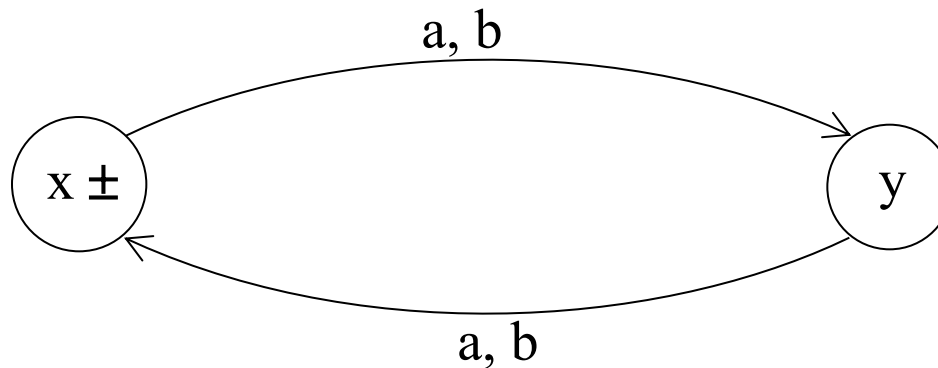
NFA equivalent to  $FA_1FA_2$

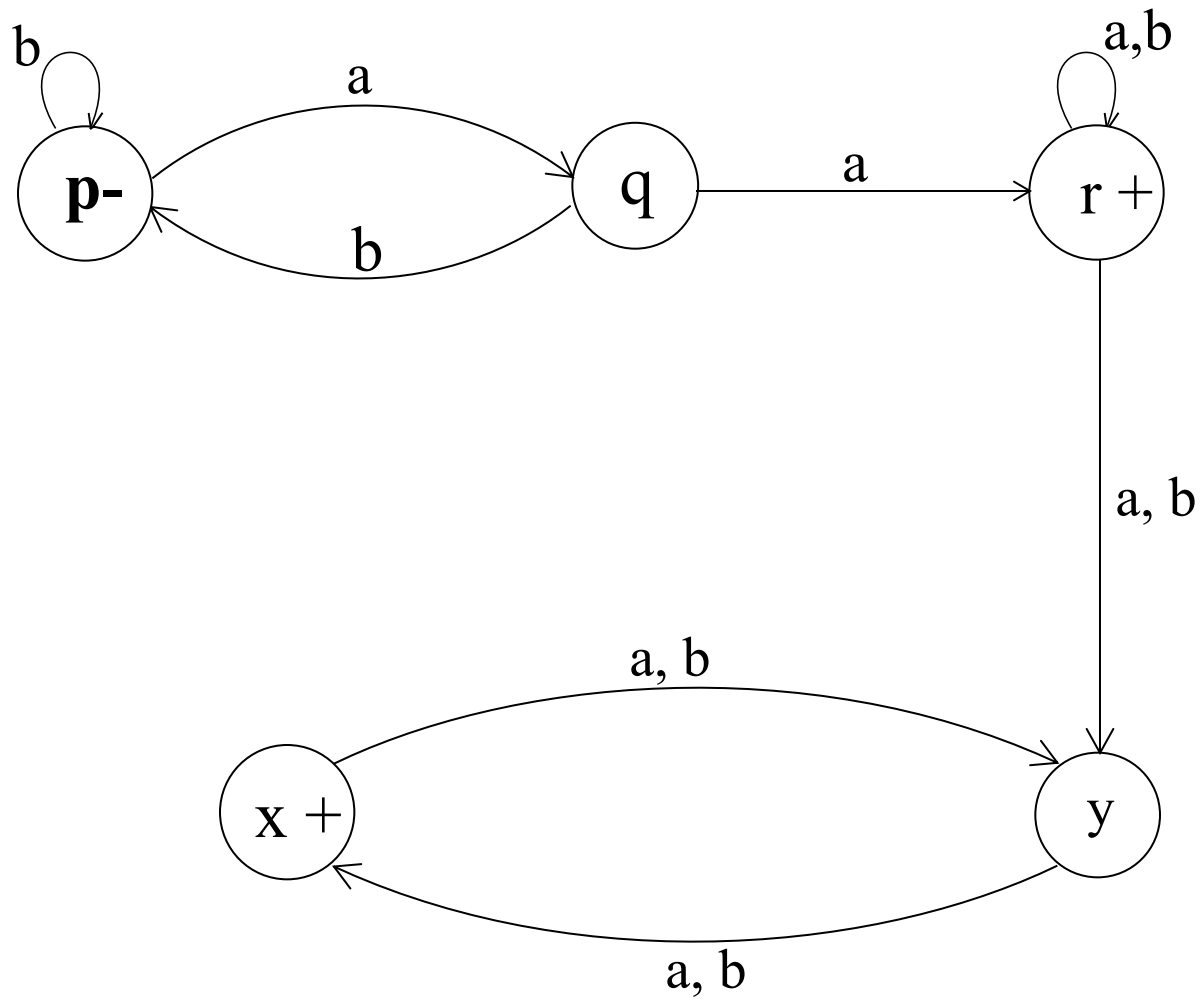
# Example ( $FA_2$ accepts Null string)

- $FA_1$



- $FA_2$

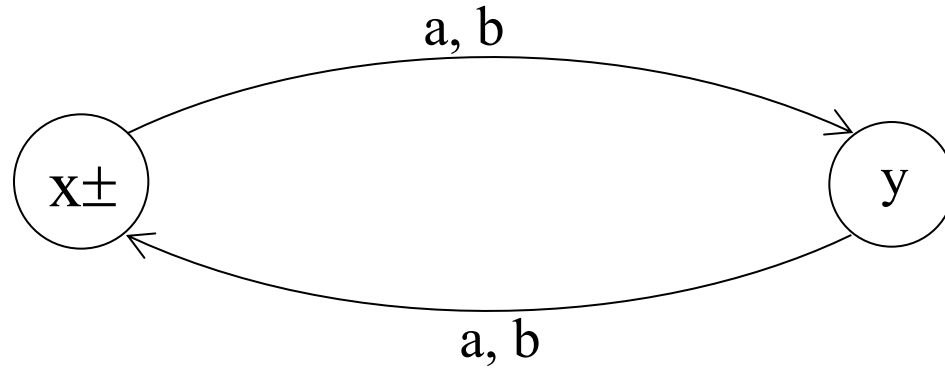




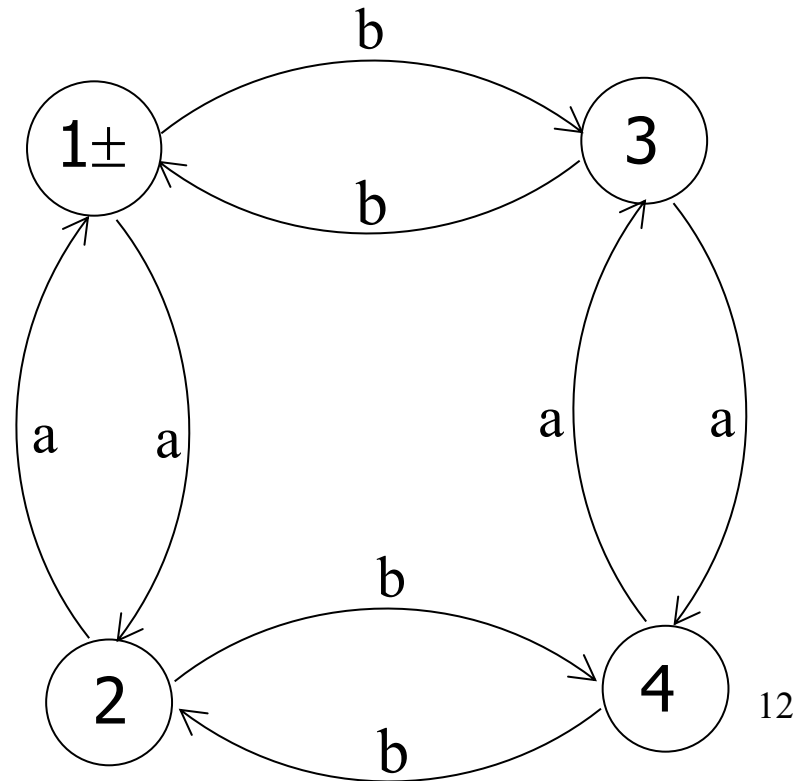
NFA equivalent to  $FA_1FA_2$

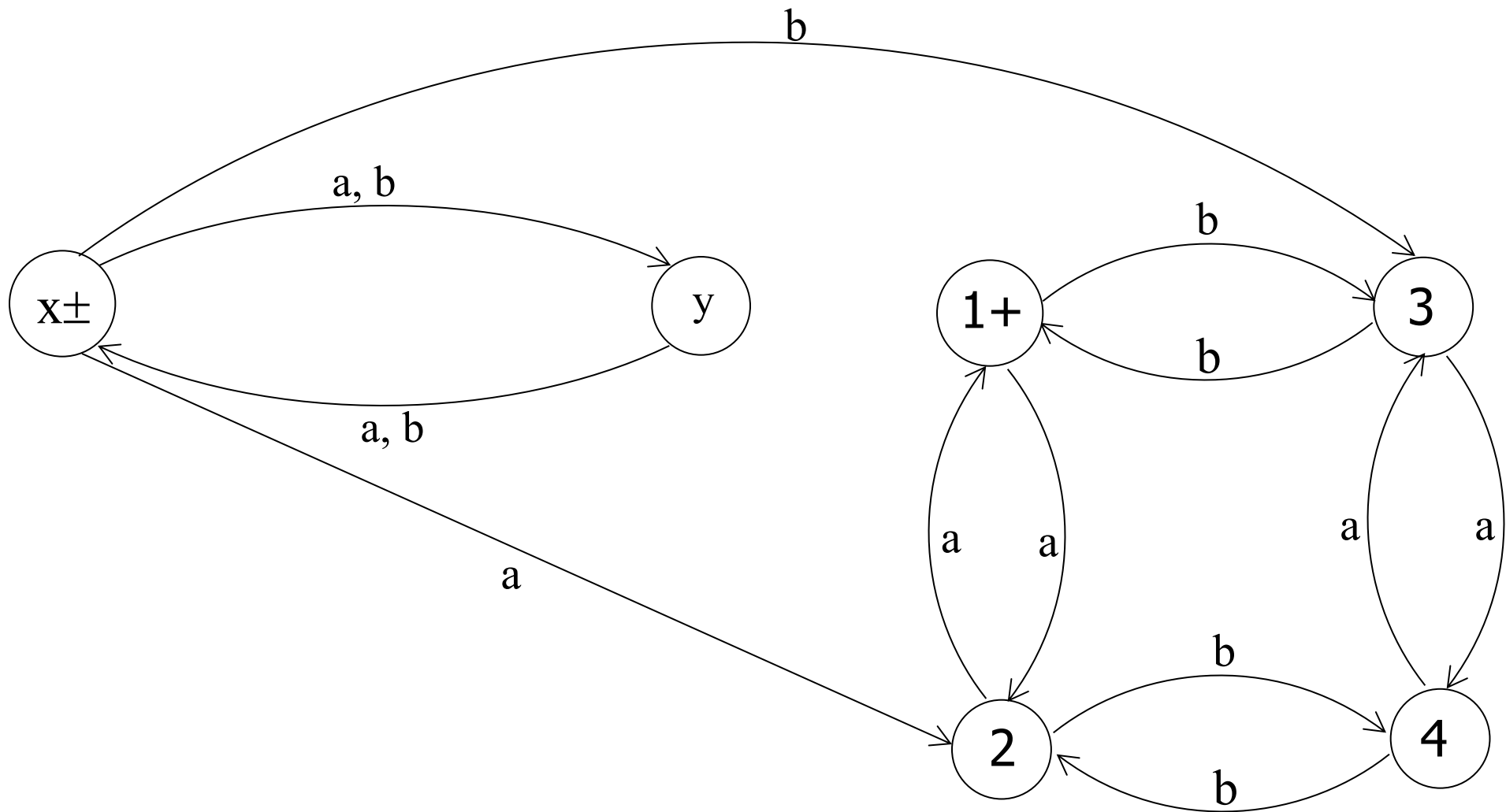
# Example (Both FAs accept Null string)

- $FA_1$



- $FA_2$





NFA equivalent to  $FA_1FA_2$

# NFA corresponding to the Closure of an FA

Apparently, it seems that since closure of an FA accepts the Null string, so the required NFA may be obtained considering the initial state of given FA to be final as well, but this may allow the unwanted string to be accepted as well. For example, an FA, with two states, accepting the language of strings, defined over  $\Sigma = \{a, b\}$ , **ending in a**, will accept all unwanted strings, if the initial state is supposed to be final as well.

# NFA for the Closure of an FA

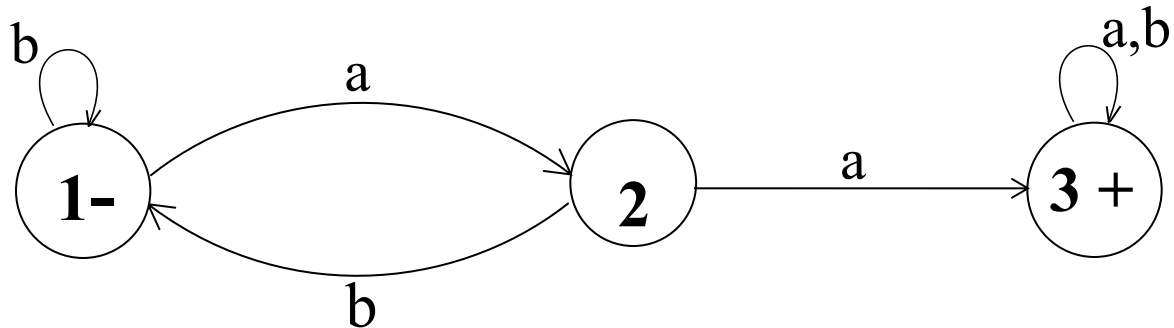
## Continued ...

### **Method:**

Thus, to accommodate this situation, introduce an initial state which should be final as well (so that the Null string is accepted) and connect it with the states originally connected with the old start state with the same transitions as the old start state, then remove the -ve sign of old start state. Introduce new transitions, for each letter, at each of the final states (including new final state) with those connected with the old start state. This creates non-determinism and hence results in the required NFA.

# Example

Consider the following FA

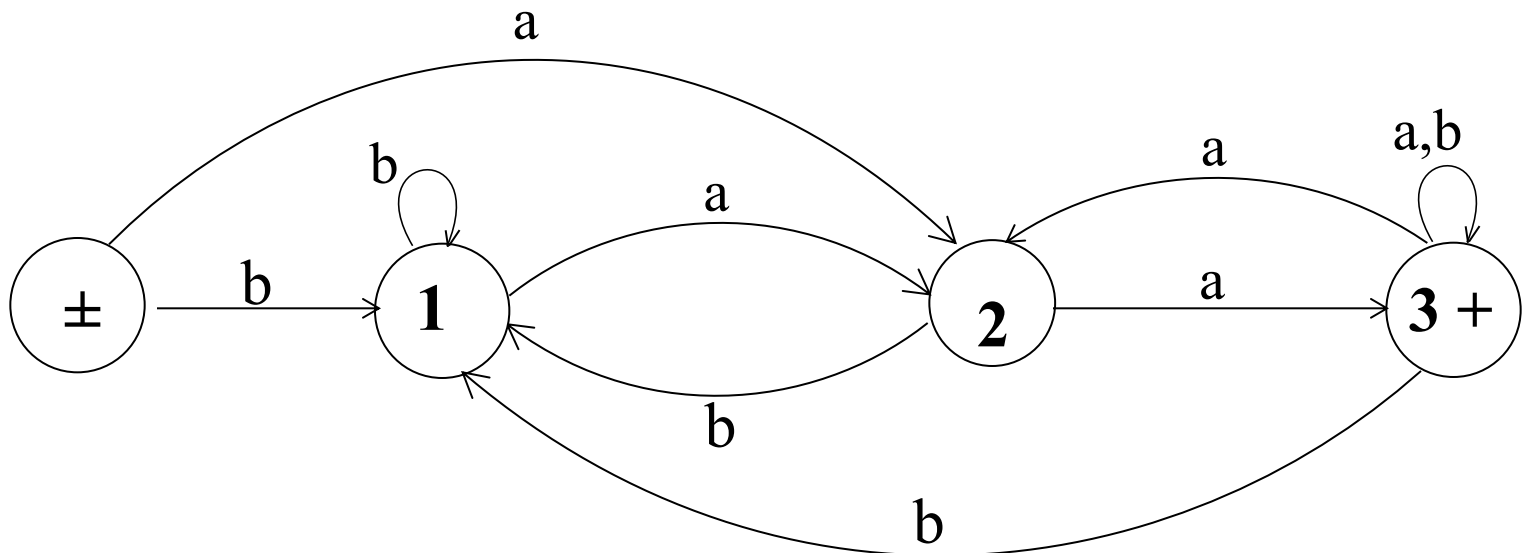


It may be observed that the FA\* accepts only the additional string which is the Null string.



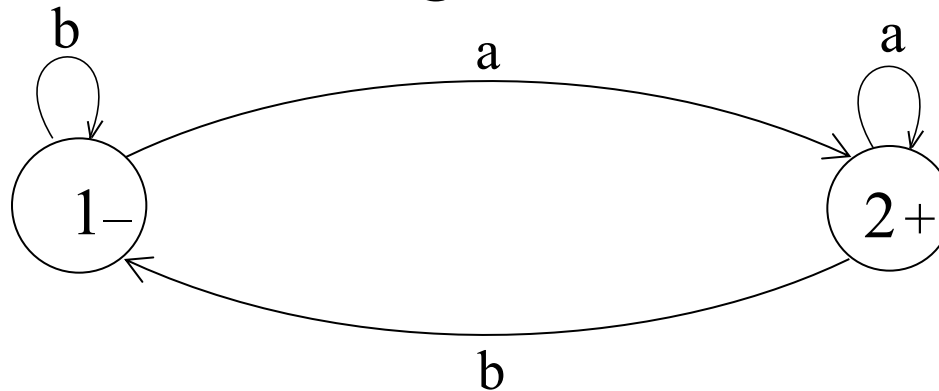
# Example Continued ...

Considering the state 1 to be final as well, will allow the unwanted strings be accepted as well. Hence the required NFA is constructed introducing the new initial state, shown below.



# Example

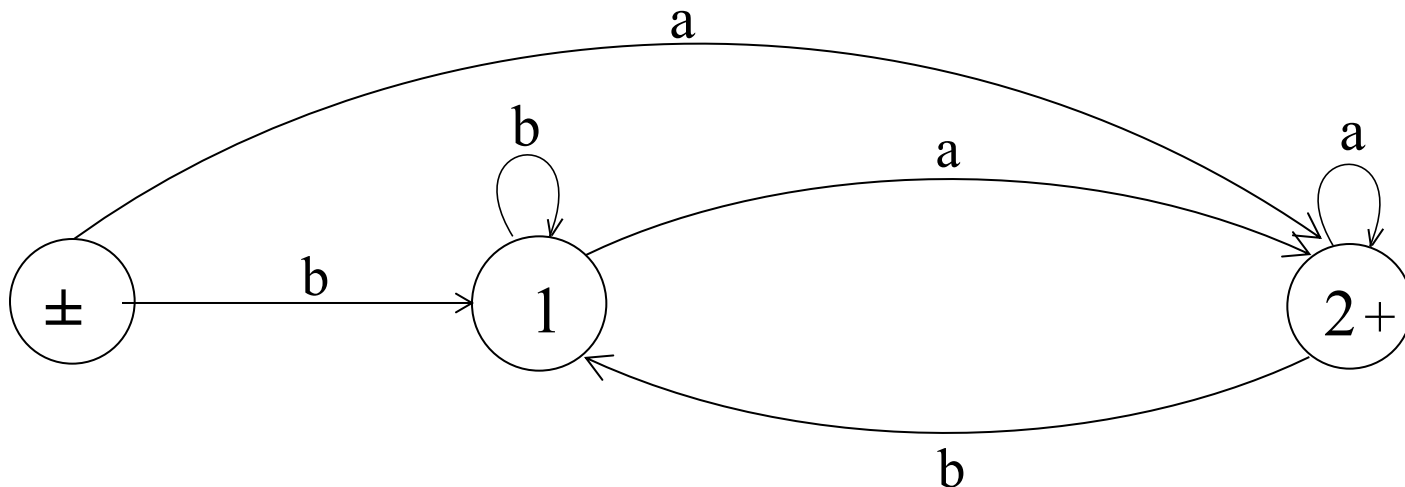
Consider the following FA



It may be observed that the FA<sup>\*</sup> accepts only the additional string which is the Null string

# Example Continued ...

As observed in the previous example the required NFA can be constructed only if the new initial state is introduced as shown below.



# Summing Up

NFA corresponding to union of  
FAs, example, NFA corresponding to  
concatenation of FAs, examples, NFA  
corresponding to closure of an  
FA, example